AI-Driven Diabetes Detection System

BUSINESS ANALYSIS 3.2

GROUP MEMBERS:

1. D MABUNDA 223975753
2. FL PHOKUNGWANA 224119966
3. PN MABASA 240397177
4. VD MATHOSI 224506501
5. E NONGQOTO 224569600
6. R NEMUTANZHELA 223377589
7. M NDLOVU 222435119

DECLARATION OF ORIGINALITY

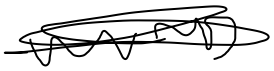
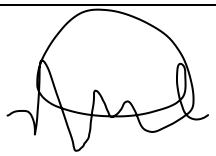
We, the undersigned, hereby declare that this project "AI Diabetes Detector" submitted is our own original work.

We confirm that:

- This project has been composed and written by us, and all sources of information, data, or code used have been properly acknowledged and cited.
- No part of this submission has been copied or plagiarized from any other group's work.
- This work has not been previously submitted, in whole or in part, for any other module at the Vaal University of Technology or any other institution.

We understand that plagiarism is both unethical and illegal, and that it is punishable by the University leadingto disciplinary action or expulsion.

INITIALS	STUDENT NUMBER	Signatute
FL PHOKUNGWANA	224119966	
PN MABASA	240397177	
VD MATHOSI	224506501	

M NDLOVU	222435119	
D MABUNDA	223975753	
E NONGQOTO	224569600	
R NEMUTANZHELA	223377589	

1. Project Background & Theme Relevance

Developing and underdeveloped regions continue to face challenges in using Fourth Industrial Revolution (4IR) technologies to address everyday community needs. In line with the theme "An AI Solution for Industries," this project proposes a Python-based artificial intelligence solution for the healthcare industry. The system will use supervised machine learning to analyse health indicators and lifestyle information to determine whether a person has diabetes or not. The system can help raise awareness and encourage people to seek appropriate healthcare. It is intended as a support tool and not as a replacement for professional medical assessment or diagnosis.

2. Problem Definition

2.1 What is the Problem?

Diabetes is a long-term health condition that affects how the body regulates blood glucose. Some people may have diabetes without recognising the symptoms, and delayed testing or care can increase the chances of serious complications. Access to screening and health information may be limited for people who face financial, geographic, or healthcare-access barriers. Diabetes risk and status are associated with a range of factors, including body mass index, blood pressure, cholesterol, physical activity, general health, and other health and demographic indicators. Considering several factors together can be difficult to do consistently without analytical support. There is therefore an opportunity to develop an accessible machine-learning tool that analyses available health indicators and produces an estimated diabetes classification to support awareness and prompt appropriate follow-up. The prediction must not be treated as a confirmed medical diagnosis.

2.2 Local Municipality and Community Benefit

An AI-driven diabetes screening support tool could be useful to community health centres, local clinics, pharmacies, and health-awareness programmes. It could analyse a set of recorded health indicators and flag cases that may need further discussion or formal testing by a qualified healthcare professional. Used responsibly, this could support health education and help direct attention toward people who may need follow-up.

The project could also demonstrate how data analysis and machine learning can be applied to public-health challenges. At community level, the system may help communicate the importance of lifestyle and health indicators associated with diabetes. Any use with real people would require appropriate privacy safeguards, careful evaluation, and clinical oversight; model predictions alone should not be used to make treatment decisions.

3. System Aims and Business Objectives

3.1 Main Aim

The main aim of this project is to develop a Python-based machine-learning application that classifies records into the two outcome categories provided by the diabetes dataset: diabetes-negative (0) and diabetes-positive (1). The model will learn from the dataset's health, lifestyle, and demographic indicators. The target column is Diabetes_binary, while the remaining 21 columns are candidate input features.

3.2 Business Objectives and Success Criteria

- Risk Classification: The system should return a binary prediction corresponding to the dataset's Diabetes_binary target: 0 (diabetes-negative) or 1 (diabetes-positive). The output must be described as a model estimate, not a medical diagnosis.
- Efficiency: Once trained, the model should return a prediction quickly enough for interactive preliminary screening. The team should measure prediction time during testing rather than assume that a particular response time has already been achieved.
- Data Use and Accessibility: The model relies only on the data provided. If we use this in the real world later, we must protect user privacy, ask easy-to-understand questions, and ensure users are guided to proper medical care.

3.3 Target Users

- Main Users: The project's intended users include healthcare-awareness programmes, community health workers, and clinics.
- Other Users: Individuals who want to see if they may have diabetes or not, or some related symptoms.

4. Machine Learning Specification & Architecture

4.1 Dataset Preparation and Feature Engineering

The supplied dataset contains 253,680 records and 22 columns. Diabetes_binary is the target variable. The remaining 21 columns are candidate model inputs. Before training, the team checked data types, missing or invalid values, class balance, duplicate records, and the dataset's coding scheme. The dataset's encoded values are interpreted using the relevant data dictionary rather than assuming that numeric codes are direct measurements.

The 21 input features in the supplied dataset are:

1. HighBP: Indicates whether the respondent reported high blood pressure.
2. HighChol: Indicates whether the respondent reported high cholesterol.
3. CholCheck: Indicates whether the respondent had a cholesterol check within the period defined by the dataset.
4. BMI: Body mass index, a measure derived from height and weight.
5. Smoker: Indicates smoking status according to the dataset's coding.
6. Stroke: Indicates whether the respondent reported having had a stroke.
7. HeartDiseaseorAttack: Indicates a reported history of coronary heart disease or heart attack.
8. PhysActivity: Indicates whether the patient reported physical activity, as defined by the dataset.
9. Fruits: Indicates fruit consumption according to the dataset's coding.
10. Veggies: Indicates vegetable consumption according to the dataset's coding.
11. HvyAlcoholConsump: Indicates heavy alcohol consumption according to the dataset's coding.
12. AnyHealthcare: Indicates whether the respondent reported having healthcare coverage or access, as defined by the dataset.
13. NoDocbcCost: Indicates whether cost prevented the respondent from seeing a doctor when needed.
14. GenHlth: Self-reported general health rating.
15. MentHlth: Number of days of poor mental health reported within the dataset's reference period.
16. PhysHlth: Number of days of poor physical health reported within the dataset's reference period.
17. DiffWalk: Indicates difficulty walking or climbing stairs.
18. Sex: Sex category encoded numerically in the dataset.
19. Age: Age category encoded numerically in the dataset, rather than a direct age in years.
20. Education: Education-level category encoded numerically.
21. Income: Income category encoded numerically.

4.2 Supervised Learning Approach and Algorithms

The project will use supervised machine learning because the dataset contains a labelled outcome, Diabetes_binary. The models will learn patterns from the input features and predict one of two classes. The dataset should be split into training and test sets, with preprocessing fitted only on the training data to avoid data leakage. Model performance should be monitored using appropriate measures such as accuracy, precision, recall, F1-score, and a confusion matrix; recall is particularly relevant when considering the consequences of missed positive cases.

- Primary Algorithm — Logistic Regression: Logistic Regression is a suitable baseline for binary classification and can provide estimated class probabilities. Its performance should be evaluated on held-out data, and the results should be interpreted cautiously because associations in the dataset do not establish causation.
- Comparison Algorithms — We will be testing all the other supervised models to see which one is the most accurate when it comes to predicting the correct result. Its performance should be compared with Logistic Regression using the same data split and evaluation measures.

5. Theoretical Extensions

5.1 Time Series Analysis

If suitable repeated measurements become available in the future, the system could be extended to examine how health indicators change over time. Time Series Analysis could help describe changes in BMI, general health, or other recorded indicators across multiple visits. The current supplied dataset is not itself a sequence of repeated measurements for each individual, so time-series analysis would require additional data.

5.2 Solution Improvement Techniques

The model could be improved through careful validation, evaluation on data from different populations, and retraining only when suitable new data becomes available. Class imbalance, changing data patterns, and differences between the dataset and the intended users should be monitored. Any improvement should be confirmed through repeatable testing and documented performance measures.

5.3 Natural Language Processing (NLP) and Speech Recognition

A future interface could allow users to answer plain-language questions or speak their responses. Speech recognition and language support could make the system easier to use for people who have difficulty with text-based forms. Such an interface

would need to explain questions clearly, protect personal information, and avoid implying that the result is a confirmed diagnosis.

5.4 Deep Learning Integration

If the project later has access to a larger, high-quality dataset , we could explore advanced AI methods (like neural networks). However, we should only use them if they prove to be better than simpler models and we can clearly explain how they work.

5.5 Conversational Chatbot / Softbot Interface

A future chatbot or softbot, for example through a web portal or messaging platform, can guide users through questions corresponding to the model's required inputs, submit the responses to the Python model, and display the estimated category. The interface should explain the limitations of the prediction and encourage users to consult a qualified healthcare professional for testing or advice. It should not provide a diagnosis or treatment recommendation.